



LAB GUIDE. SESSION 4

GOALS:

- Greedy algorithms

1. Introduction

The Board of Directors of a company on the verge of bankruptcy proposed, with the primary objective of keeping its n employees ($employee_0, employee_1, \dots, employee_{n-1}$), in a final attempt to save it:

- The creation of n new positions for those employees: $position_0, position_1, \dots, position_{n-1}$.
- It has commissioned a renowned consulting firm to calculate (based on a multitude of parameters) the productivity that each of its employees would have in each of the positions.

This study has already been delivered and is in an input file where each productivity value is in the range $[1000 .. 9999]$, and the element $[i][j]$ is the estimated productivity that employee i would have in position j .

For example, this is a file **datos4.txt**:

```
4
5960,7118,6751,1772
8927,4255,6801,6731
4078,4907,7164,7211
4336,7767,5849,5482
```

What remains to be determined is the assignment of each employee to one of the newly created positions. For this assignment, the company's CEO has established the following 3 steps:

1. An **ALGORITHM 1** that simulates assigning to each employee (by index order) the job position in which they have the highest productivity among the positions that have not been previously assigned. This assignment will result in a total productivity of $P1$ (the sum of the n assigned productivities).
2. An **ALGORITHM 2** that simulates assigning each position (by index order) to the employee who has the highest productivity among the employees who have not been previously assigned. This assignment will result in a total productivity of $P2$ (the sum of the n assigned productivities).
3. Finally, the actual assignment to be carried out is chosen based on $P = \text{maximum}(P1, P2)$.

We are given an input dataset consisting of 5 files:

- "Datos4.txt": for this case, the resulting solution is in "Resultados4.txt".
- "Datos16.txt": for this case, the resulting solution is in "Resultados16.txt".
- "Datos64.txt": for this case, the resulting solution is in "Resultados64.txt".
- "Datos256.txt": for this case, the resulting solution is in "Resultados256.txt".
- "Datos1024.txt": for this case, the resulting solution is in "Resultados1024.txt".

YOU ARE REQUESTED TO:

Solve the proposed problem creating an **Assignment.java** class.

Execute the class by passing each of the provided file names as a parameter to verify the correct functioning of the implemented algorithm.

Create an **AssignmentTimes.java** class that, after randomly generating an $n \times n$ productivity matrix, calculates the execution time of the assignment algorithm:

*The time should be in milliseconds, with the optimizer enabled, and marked as **TLE** (Time Limit Exceeded) when it takes longer than one minute.*

n	$time$
1024	
2048	
4096	
...	
until it takes more than a minute	

Create an **AssignmentTests.java** class that contains 5 test cases (5 methods), one for each input file, to verify that the result is as expected. The test cases must run without requiring any modifications to your source code. Otherwise, the exercise will be graded as a 0.

- Reason whether the obtained times align with the complexity of the algorithm (using the formulas).
- Do you think the proposed algorithms (ALGORITHM 1 and ALGORITHM 2) are greedy algorithms? Why?
- Do you think the problem would allow for a divide-and-conquer solution? Why?
- Does the ASSIGNMENT proposed by the CEO and currently implemented achieve the optimal solution (the highest possible total productivity)? Why?

2. Work to be done

- An `algstudent.s4` **package** in your course project. The content of the package should be the Java files used and created during this session.
- A `session4.pdf` **document** using the course template (the document should be included in the same package as the code files). You should create one activity each time you find a "YOU ARE REQUESTED TO" instruction.

Deadline: The lab must be submitted on time and in accordance with the instructions provided by the lab instructor.

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